

Tamjid Al Rahat

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Education

- **University of California Los Angeles (UCLA)**, California 2024
Ph.D. in Computer Engineering
CGPA: 4.0/4.0
- **University of Virginia**, Charlottesville, Virginia 2020
Master of Computer Science
CGPA: 3.90/4.0.
- **Bangladesh University of Engineering and Technology**, Bangladesh 2016
B.Sc. in Computer Science and Engineering
CGPA: 3.84/4.0

Research Interest

AI/ML for security, systems & software security, vulnerability analysis, formal methods, program analysis, automated reasoning

Experience

- **Amazon Web Services (AWS)**, Cupertino, California 07.2024-Present
- *Applied Scientist, AWS Proactive Security*
- **University of California, Los Angeles (UCLA)**, California 09.2022-06.2024
- *Graduate Research Assistant, UCLA Security Lab*
- **University of Virginia**, Charlottesville, Virginia 08.2018-08.2022
- *Graduate Research Assistant, Computer Science*
- **United International University**, Bangladesh 06.2016-07.2018
- *Lecturer, Computer Science and Engineering*
- **Stochastic Logic Limited**, Bangladesh 05.2016-07.2018
- *Quantitative Software Developer*

Publications

- **ACM CCS 2024**: T. Al Rahat, Y. Feng, and Y. Tian, “AuthSaber: Automated Safety Verification of OpenID Connect Programs”. 31th ACM Conference on Computer and Communications Security.
- **ACM CCS 2022**: T. Al Rahat, Y. Feng, and Y. Tian, “Cerberus: Query-driven Scalable Vulnerability Detection in OAuth Service Provider Implementations”. 29th ACM Conference on Computer and Communications Security.
- **IEEE TPS 2022 [Best Paper Award]**: T. Saha, T. Al Rahat, N. Aaraj, Y. Tian, N. Jha, “ML-FEED: Machine Learning Framework for Efficient Exploit Detection”. IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications.
- **ACM WPES 2022**: T. Al Rahat, M. Long, and Y. Tian, “Is Your Policy Compliant? A Deep Learning-based Empirical Study of Privacy Policies’ Compliance with GDPR”. 21st Workshop on Privacy in

the Electronic Society.

- **IEEE/ACM ASE 2019:** T. Al Rahat, Y. Feng, and Y. Tian, "OAuthLint: An Empirical Study on OAuth Bugs in Android Applications". 34th IEEE/ACM Conference on Automated Software Engineering.
- **ADC 2018:** T. Al Rahat, A. Arman, M. E. Ali, "Maximizing Reverse k-Nearest Neighbors for Trajectories". 29th Australasian Database Conference.
- **VEHITS 2020:** T. Le, I. ElSayed-Aly, W. Jin, S. Ryu, G. Verrier, T. Al Rahat, B. Park, and Y. Tian, "Evaluating the Dedicated Short-range Communication for Connected Vehicles against Network Security Attacks". 6th Int. Conference on Vehicle Technology and Intelligent Transport Systems .

Selected Research Projects

- **Scalable security analysis of large-scale cloud applications**
 - Developed a novel security analysis approach to automatically detect vulnerable implementations of authorization protocols in large cloud-based applications.
 - Identified critical security vulnerabilities in many popular service providers, including **Google**, **GoFundMe** and 10+ open source projects, and discovered five new **CVEs** (published in NIST's National Vulnerability Database).
- **Automated large-scale analysis of privacy law violations**
 - Built a new policy dataset and developed a *Deep Active Learning*-based solution to automatically measure compliance of data practices with privacy laws like *GDPR*.
- **Automated and adaptive security analysis of next-generation protocols**
 - Proposed an automated and adaptive solution for security analysis in IoT Matter protocols.
- **Cross-platform vulnerability analysis with Transfer Learning**
 - Developed a *Transfer Learning*-based AI model to detect common vulnerabilities across multiple platforms and programming languages.

Selected Other Projects

- **Vulnerable program-flow extraction from Java apps**
 - Developed a new tool that automatically extracts attack entities from CVE dataset and finds vulnerable instruction sequences from a given Java program.
- **Automatic detection of security flaws in Python web apps**
 - Developed a tainted data-flow analysis tool that identified 700+ *security flaws* in 450 open-source Python web apps in Github.
- **Pneumonia detection in X-rays**
 - Implemented a *ResNet architecture* to detect the pneumonia affected areas in chest x-rays with 77.5% Intersection over Union (IoU).
- **Acoustic-based occupancy detection**
 - Extracted audio features from data collected by a voice assistant prototype and applied Random Forest to achieve 97% accuracy for occupancy detection.
- **Lung Cancer detection in CT scans**
 - Used *Transfer Learning* to detect lung nodules in 3D CT scans from LIDC/IDRI dataset.
- **CNN-based price prediction from product images**
 - Used *Transfer Learning* to extract visual features from Amazon product images and predict their

prices.

Technical Skills

- **Programming Language:** Python, Java, C/C++, Racket, Intel x86 Assembly, Matlab
- **Program analysis tools:** Wala, Soot, Ghidra, Pysa, Clang, Z3, Tamarin-prover, Rosette, FlowDroid
- **Machine learning tools:** Tensorflow, Keras, PyTorch, Scikit-learn
- **Web:** PHP, Javascript, MySQL, SQLite, Oracle
- **Scripting:** $\text{\LaTeX}$, Bash, Html, Lisp

Awards and Grants

- **Google Research Paper Reward**, 2023.
- **Qualcomm Innovation Fellowship**, Finalist, 2023.
- **Google Vulnerability Reward**, 2022 (Won 5000\$ for reporting security critical bugs).
- **Vulnerability Research Grant, Google**, 2022 (For security research on Google Identity Services).
- **Graduate Education Fellowship**, UCLA, 2022-2023.
- **UVA Computer Science Fellowship**, University of Virginia, 2018-2019.
- **Bug Bounty**, GoFundMe, 2019.
- **Travel Grant**, Amazon's Graduate Research Symposium, 2019.
- **Dean's List Award**, Bangladesh University of Engineering and Technology, 2012-2016.
- **University Merit Scholarship**, Bangladesh University of Engineering and Technology, 2012-2016.
- **General Scholarship**, Secondary School Certificate Examination, 2008.

Academic Service

- **Session Chair:** ACM CCS '22, ACM WPES '22
- **Technical Program Committee:** ACM CCS '22 (posters), IEEE S&P '23 (posters)
- **Peer Reviewer:** CSF '21, ICSTE '22, JSS '22-23
- **External Peer Reviewer:** S&P '21-23, CCS '21-23, Usenix '21-22, WWW '22, SenSys '21-22

Media Coverage

- *High-Severity Bug Reported in Google's OAuth Client Library for Java*, The Hacker News, May 2022.